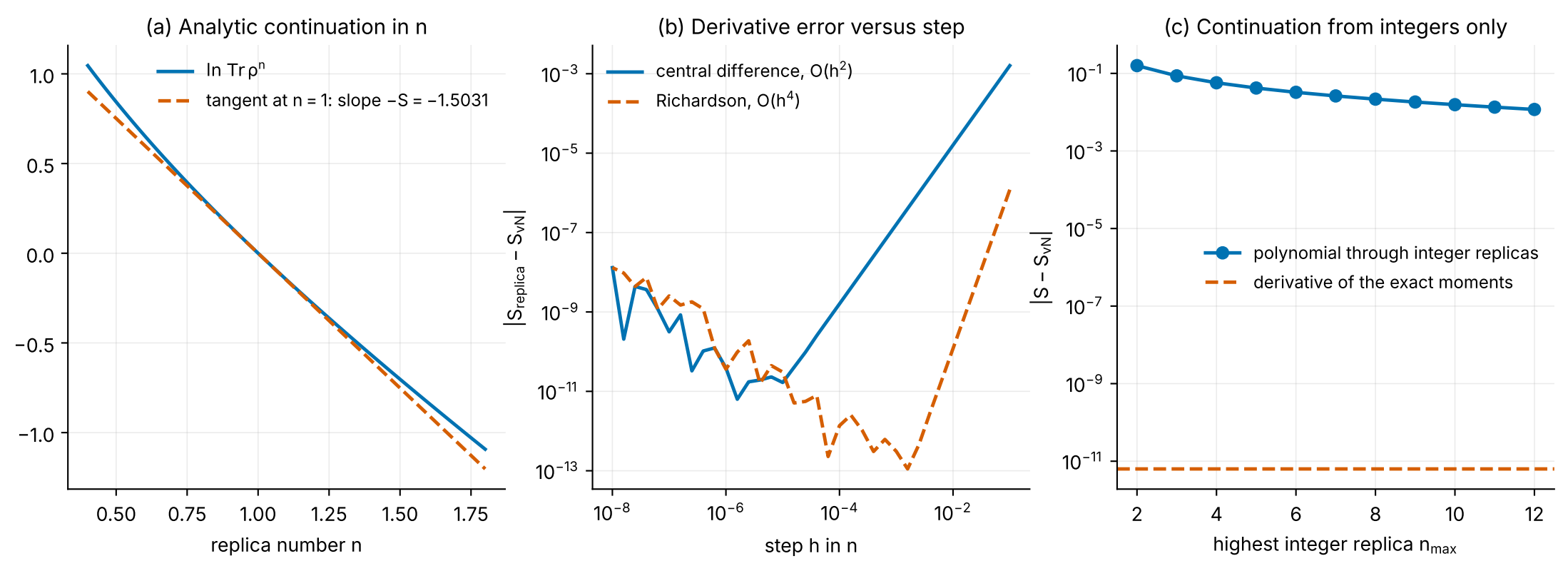
(a) Analytic continuation in n
ln Tr ρ^n
tangent at n = 1: slope −S = −1.5031
1.0
0.5
0.0
−0.5
−1.0
0.50
0.75
1.00
1.25
1.50
1.75
replica number n
(b) Derivative error versus step
central difference, O(h^2)
Richardson, O(h^4)
|S_replica − S_vN|
10^−3
10^−5
10^−7
10^−9
10^−11
10^−13
10^−8
10^−6
10^−4
10^−2
step h in n
(c) Continuation from integers only
polynomial through integer replicas
derivative of the exact moments
|S − S_vN|
10^−1
10^−3
10^−5
10^−7
10^−9
10^−11
2
4
6
8
10
12
highest integer replica n_max